Cover letter

12 October 2026

Editors, *Journal of Intelligent Manufacturing*

Dear Editors,

Please consider our manuscript, "Reinforcement Learning-Based Computer Numerical Control Motion Planning with Adaptive Preview and Jerk-Constrained Action Projection", for publication as a research article.

Our manuscript presents an offline learned CNC planner that jointly selects steering, feedrate and preview. A trained policy and planar kinematic projection generate the command trajectory. Startup and braking refinements preserve intermediate commands, and the final plan is checked against the geometric and kinematic constraints.

The simulation study compares four selected complete-path cases, comprising two development programs and two enlarged synthetic cases, under common evaluation conditions. Analysis of the command trajectories relates traversal time to path geometry and speed variation.

The method combines learned decisions with explicit trajectory constraints for CNC motion planning. This application of artificial intelligence to manufacturing-system control fits the scope of the Journal of Intelligent Manufacturing.

Sincerely,

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